

MAINS MASTER PROGRAM (MMP) 2024

ENVIRONMENT-1

AIR POLLUTION ISSUES

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2. SYLLABUS

Conservation, environmental pollution and degradation, environmental impact assessment

3. PYQS

1. **[2023]:** Discuss the consequence of climate change on the food security in tropical countries **[GS1, 10 marks, 150 words]**
2. **[2023]:** Why is the world today confronted with a crisis of availability and access to freshwater resources? **[GS1, 10 marks, 150 words]**
3. **[2023]:** Identify and discuss the factors responsible for diversity of natural vegetation in India. Assess the significance of wildlife sanctuaries in rain forest regions of India. **[GS1, 10 marks, 150 words]**
4. **[2023]:** What is oil pollution? What are its impact on marine ecosystem? In what way is oil pollution particularly harmful for a country like India **[GS3, 10 marks, 150 words]**
5. **[2023]:** Comment on the National Wetland Conservation Program initiated by Government of India and name a few of India's wetland of international importance included in the Ramsar sites. **[GS3, 15 marks, 250 words]**
6. **[2023]:** The IPCC has predicted a global sea level rise of about one metre by 2100 AD. What would be its impact on India and the other countries in the Indian Ocean region **[GS3, 15 marks, 250 words]**
7. **[2022]:** Examine the potential of wind energy in India and explain the reasons for their limited spatial spread **[GS1, 10 marks, 150 words]**
8. **[2022]:** Discuss in details about the photochemical smog emphasizing its formation, effects and mitigation. Explain the 1999 Gothenburg Protocol **[GS3, 10 marks, 150 words]**
9. **[2022]:** Discuss global warming and mention its effects on the global climate. Explain the control measures to bring down the level of greenhouse gases which cause global warming, in the light of Kyoto Protocol, 1997 **[GS3, 15 marks, 250 words]**
10. **[2022]:** Explain the causes and effects of coastal erosion in India. What are the available coastal management techniques for combating these hazards **[GS3, 15 marks, 250 words]**
11. **[2021]:** What are the environmental implications of the reclamation of water bodies into urban land use? Explain with examples **[GS1, 10 marks, 150 words]**
12. **[2021]:** How do melting of the Arctic ice and glaciers of the Antarctic differently able to affect the weather patterns and human activities on the Earth? Explain **[GS1, 15 marks, 250 words]**
13. **[2021]:** Explain the purpose of Green Grid Initiative launched at the World Leaders Summit of the COP26 UN Climate Change Conference in Glasgow in Nov 2021. When was the idea first floated in the International Solar Alliance (ISA)? **[GS3, 10 marks, 150 words]**
14. **[2021]:** Describe the key points of the revised Global Air Quality Guidelines (AQGs) recently released by the WHO. How are these different from its last update in 2005? What changes in India's National Clean Air Program (NCAP) are required to achieve these revised standards? **[GS3, 10 marks, 150 words]**
15. **[2021]:** Describe the major outcomes of the 26th session of COP to the UNFCCC. What are the commitments made in this conference **[GS3, 15 marks, 250 words]**
16. **[2020]:** The process of desertification doesn't have climatic boundaries. Justify with examples **[GS1, 10 marks, 150 words]**
17. **[2020]:** Examine the status of forest resources of India and its resultant impact on climate change **[GS1, 15 marks, 250 words]**

18. **[2020]**: How does the draft Environmental Impact Assessment (EIA) Notification, 2020 differ from the existing EIA notification, 2006? [GS3, 10 marks, 150 words]
19. **[2020]**: What are the salient features of the Jal Shakti Abhiyan launched by the Government of India for water conservation and water security? [GS3, 10 marks, 150 words]
20. **[2020]**: What are the key features of the National Clean Air Program (NCAP) initiated by the Government of India? [GS3, 15 marks, 250 words]
21. **[2019]**: Assess the impact of global warming on the coral life system with examples [GS1, 10 marks, 150 words]
22. **[2019]**: Discuss the causes of depletion of mangroves and explain their importance in maintaining coastal ecology [GS1, 10 marks, 150 words]
23. **[2019]**: What is water stress? How and why does it differ regionally in India [GS1, 15 marks, 250 words]
24. **[2019]**: How can the mountain ecosystem be restored from the negative impact of development initiative and tourism? [GS1, 15 marks, 250 words]
25. **[2019]**: Coastal sand mining, whether legal or illegal, poses one of the biggest threats to our environment. Analyze the impact of sand mining along the Indian coasts, citing specific examples. [GS3, 10 marks, 150 words]
26. **[2019]**: Define the concept of carrying capacity of an ecosystem as relevant to an environment. Explain how understanding this concept is vital while planning for sustainable development of a region. [GS3, 15 marks, 250 words]
27. **[2018]**: What are the consequences of spreading of 'Dead Zones' on marine ecosystem? [GS1, 10 marks, 150 words]
28. **[2018]**: The ideal solution of depleting ground water resources in India is water harvesting system. How can it be made effective in urban areas. [GS1, 15 marks, 250 words]
29. **[2018]**: What are the impediments in disposing the huge quantities of discarded solid wastes which are continuously being generated? How do we remove safely the toxic wastes that have been accumulating in our habitable environment [GS3, 10 marks, 150 words]
30. **[2018]**: What is wetland? Explain the Ramsar concept of 'wise use' in the context of wetland conservation. Cite two examples of Ramsar site in India [GS3, 10 marks, 150 words]
31. **[2018]**: How does biodiversity vary in India? How is the Biological Diversity Act, 2002 helpful in conservation of flora and fauna [GS3, 15 marks, 250 words]
32. **[2017]**: "In spite of adverse environmental impact, coal mining is still inevitable for development." Discuss [GS1, 10 marks, 150 words]
33. **[2017]**: How does the cryosphere affect the global climate? [GS1, 10 marks, 150 words]
34. **[2017]**: Not many years ago, river linking was a concept but it is becoming a reality in the country. Discuss the advantages of river linking and its possible impact on the environment [GS3, 10 marks, 150 words]
35. **[2017]**: 'Climate Change' is a global problem. How India will be affected by climate change? How Himalayas and coastal states of India will be affected by climate change? [GS3, 10 marks, 150 words]
36. **[2016]**: 'Rehabilitation of human settlements is one of the important environmental impacts which always attracts controversy while planning major projects. Discuss the measures suggested for mitigation of this impact while proposing major developmental projects' [GS3, 12.5 marks, 200 words]
37. **[2015]**: Mumbai, Delhi, and Kolkata are the three mega cities of the country but the air pollution is much more serious problem in Delhi as compared to the other two. Why is this so. [GS1, 12.5 marks, 200 words]
38. **[2015]**: India is well endowed with fresh water resources. Critically examine why it still suffers from water scarcity. [GS1, 12.5 marks, 200 words]
39. **[2015]**: The states of Jammu and Kashmir, Himachal Pradesh, and Uttarakhand are reaching the limits of their ecological carrying capacity due to tourism. Critically Evaluate [GS1, 12.5 marks, 200 words]

40. **[2015]:** Discuss the Namami Gange and National Mission for Clean Ganga (NMCG) Programmes and causes of mixed results from the previous schemes. What quantum leaps can help the river Ganga better than incremental inputs? **[GS3, 12.5 marks, 200 words]**
41. **[2014]:** Bring out the relationship between Shrinking Himalayan glaciers and the symptoms of climate change in the Indian sub-continent **[GS1, 10 marks, 150 words]**
42. **[2014]:** Should the pursuit of carbon credits and clean development mechanisms set up under UNFCCC be maintained even though there has been a massive slide in the value of carbon credit? Discuss with respect to India's energy needs for economic growth. **[GS3, 10 marks, 150 words]**
43. **[2014]:** Environmental Impact Assessment are increasingly undertaken before a project is cleared by the Government. Discuss the environmental impact of coal-fired thermal power plants located at coal pitheads. **[GS3, 10 marks, 150 words]**
44. **[2013]:** What are the consequences of illegal mining? Discuss the Ministry of Environment and Forests' concept of GO and NO-GO zones for coal mining sector. **[GS3, 10 marks, 200 words]**
45. **[2013]:** Enumerate the National Water Policy of India. Taking River Ganges as an example, discuss the strategies which may be adopted for river water pollution control and management. What are the legal provisions of management and handling of hazardous wastes in India? **[GS3, 10 marks, 200 words]**
46. **[2012]:** In the context of growing demands for the ban of endosulfan in the country, critically examine the issues involved? What, in your view, should be done in the matter?

4. AIR POLLUTION AND RELATED ISSUES

[2022]: Discuss in detail about the photochemical smog emphasizing its formation, effects and mitigation. Explain the 1999 Gothenburg Protocol **[GS3, 10 marks, 150 words]**

[2021]: Describe the key points of the revised Global Air Quality Guidelines (AQGs) recently released by the WHO. How are these different from its last update in 2005? What changes in India's National Clean Air Program (NCAP) are required to achieve these revised standards? **[GS3, 10 marks, 150 words]**

[2020]: What are the key features of the National Clean Air Program (NCAP) initiated by the Government of India? **[GS3, 15 marks, 250 words]**

[2015]: Mumbai, Delhi, and Kolkata are the three mega cities of the country, but the air pollution is much more serious problem in Delhi as compared to the other two. Why is this so. **[GS1, 12.5 marks, 200 words]**

1) AIR POLLUTION SITUATION IN INDIA AND ITS IMPACT

- **Introduction: Current Pollution situation in India:**
 - » As per the World Air Quality Report by IQ Air, India is the third most polluted country in the world, and it has 9 out of the 10 most polluted cities in the world.
- **Major sources** of air pollution in India include Transportation, biomass burning for cooking, Electricity generation, industry, construction, waste burning, and episodic agricultural burning.
- **Impacts**
 - i. **Health** - As per the Air Quality Life Index Report 2023, in North India, PM2.5 shortens lives by eight years.
 - » It can cause mental health issues, Alzheimer's and loss of vision. Women chronically exposed to particulate air pollution are more likely to have underweight babies and preterm birth.
 - ii. **Economic loss:** Studies have shown that overall economic cost of air pollution is around **7% of the GDP (Rs 20 lakh crores)**. It happens due to deaths, disabilities, reduction in labor productivity, Agricultural losses (farm labor productivity, reduction in crop), tourism revenue loss etc
 - iii. **Environmental Impact:**
 - » Dust and particulate matter may reduce the yields of solar power plants.
 - » Some air pollutants like methane, ozone etc are also green house gases and may cause global warming.
 - » Air pollution may also cause acid rain (NO₂, SO₂) etc which may damage forests, crops, and aquatic ecosystem.
 - iv. **Hampers Aesthetics of the city:** Haze and smog may hamper the aesthetics of the city.
 - v. **Ozone Layer depletion:** Some air pollutants like CFCs are also ozone depleting and this may increase the ultraviolet rays reaching earth.
- **Key challenges in fighting Air Pollution:**
 - » **India's Economic growth is built on fossil fuels:** Coal, Oil, and Natural Gas account for roughly 75% of our power generation and 97% of road transport.
 - » **Increasing Population, and rapid urbanization and industrialization:** It is leading to increased demand for energy which is leading to more burning of fossil fuels.
 - » **Agriculture Practices** like stubble burning, overuse of fertilizers etc. are difficult to control and these are non-point sources.
 - » **Lack of Infrastructure and Technology:** For e.g. poor conditions of public transport leads to increased dependency on private vehicles; Similarly, outdated industrial practices in thermal power plants and other industries also increases the chances of pollution.
 - » **Solid Fuel dependency for cooking in rural areas** -> firewood or dung cake -> Drudgery; Health Impact.
 - For e.g. As per the latest World Air Quality Report by IQAir, **Begusarai** in eastern Bihar is the most polluted city in the world. Experts guess that solid fuel burning may be a major reason for this.
 - » **Lack of resources** (including human resource) with local bodies and pollution control authorities: This leads to poor monitoring and compliance of the pollution norms.

- **Steps Taken:**
 - » NCAP
 - » Promotion of Renewable Energy
 - » Stricter emission standards under BS-VI
 - » Increased emphasis on improving public transport.
 - » PM-Ujjwala
 - » Mission LIFE - to bring behavioural changes among people to fight air pollution
 - » Promotion of Nano-Urea and Nano-DAP.
- **Way Forward**
 - » **Legal and Policy Initiatives:**
 - **Effective implementation of National Clean Air Program:**
 - **Phase out of existing coal, oil and gas infrastructure**
 - **Closure of old coal-fired power plants** can yield health benefits that exceed the value of electricity generated.
 - **Phase out dates for diesel, petrol and gas cars -> Promote electric vehicles**
 - **Transition to affordable and carbon neutral transport -> Strengthening of public transport, good walking and cycling infrastructure.**
 - **Transition to renewable energy.**
 - Share of renewable energy in India has risen dramatically to around 100 GW in Aug 2021, but there is still a long way to go.
 - **Healthy Way to Cook:** Analyze the shortcomings of PM Ujjwala Yojna and bring changes to ensure 100% coverage and utilization.
 - » **Deal with issue of stubble burning:** PUSA decomposer (use of biotechnology); Awareness; Higher subsidy for machinery; Crop diversification etc.
 - » **Expand country's Air Quality Monitoring Network** - A number of new, low cost monitors have entered service, that capture readings for not only PM2.5 and 10 but also other gases like NO2, SO2, methane, VoCs etc.
 - » **Strengthen CPCB and other SPCBs** to effectively implement the air pollution norms.
 - » **Revise Ambient Air Quality Standards** to bring them to **WHO levels** and **implement them** without exceptions
 - » **Pollution should become a mainstream issue:** There must be a wide-scale citizen mobilization for awareness and education, which can influence lifestyle choices over decades. This played an important role in Los Angeles in the 1960s and the Mexico City in 1990s and 2000s.
 - This will also make pollution an election issue and increase political will to fight pollution.
- **Conclusion**
 - » China was once amongst the most polluted countries in the world. But consistent efforts in the field of prioritizing zero-emissions transport, staggered use of Internal Combustion engines, promotion of Electric vehicles etc. while continuing to grow economically, shows that air pollution can be clamped down without impeding economic growth.

2) COAL FIRED POWER PLANTS AND ASSOCIATED ISSUES

- **Why in news?**
 - » Only 5% of India's coal-based thermal power capacity meets SO₂ emission norms: CSE report (June 2023)
- **Introduction:**
 - » Coal is the most important and abundant fossil fuel in India. It accounts for 55% of India's energy needs. Infact, India's industrial heritage was built upon Indigenous coal.
- **Environmental Impacts of Coal Based Thermal Power Plants:**
 - » **Air Pollution:** Burning of coal produces air pollutants like NO₂, SO₂, CO, PM, Mercury etc. which are primary air pollutants in the world.
 - » **Acid Rain:** Pollutants like NO₂, SO₂ etc. contribute to acid rain. It can harm forests, aquatic ecosystems, and buildings and it can also lead to soil and water acidification.
 - » **Improper management of fly ash** also leads to air, water and oil pollution.
 - » **Climate Change due to global warming** caused by emission of greenhouse gases. This hampers achievement of Paris Agreement targets.
 - » **Excessive water Extraction:** Coal based thermal power plants require huge quantities of water which is often drawn from nearly rivers, lakes, or groundwater sources.
 - » **Water Pollution:** Leakage of heavy metals and acids from the exposed coal seams may cause water pollution. They can leach into the groundwater and nearby waterbodies, posing risks to drinking water source and health of aquatic ecosystem. In addition, the release of warm water from the thermal power plant also causes thermal pollution.
 - » **Other issues created by mining of coals** such as land degradation, pollution during transportation etc.
- **Emission Norms:**
 - » The MoEF&CC had notified the emission norms for coal-based power plants in Dec 2015.
 - » In 2021, **MoEF&CC divided the power plants on the basis of distance from polluted cities** to enforce deadlines and extended the **deadlines**.
 - i. **Category A** - coal based power plants within 10 kms radius of NCR and of cities with million+ population. (deadline changed from 31st Dec 2022 to 31st Dec 2024)
 - ii. **Category B** - power plants within 10 kms radius of critically polluted areas or non-attainment cities. (deadline changed from 31st Dec 2023 to 31st Dec 2025)
 - iii. **Category C** - remaining plants throughout the country. (deadline changed from 31st Dec 2024 to 31st Dec 2026)
 - This has the longest deadline and most of the country's coal based power plants fall in this category.
 - » An analysis by the Centre for Research on Energy and Clean Air (CREA) has found less than 8% of India's coal-based power plants have installed the SO₂ emission reduction technology recommended by MoEF&CC.
- **Why moving away from coal based power plants may be difficult?**
 - » Very large dependency -> 75% of India's annual power output.

- » **Phasing in renewable energy sources and phasing out conventional sources rapidly** may lead to instability in the electricity grid which may potentially cause blackout.
- » **Political Economy Risk:** Aggressive early retirement of coal based capacity, without detailed analyses, could result in real or perceived electricity shortage in some states.
- **The above challenges can be dealt in the following ways:**
 - » **Steps at thermal power plant level:**
 - Close very old power plants which can't be upgraded
 - Implementing advanced pollution control technologies and strict regulation. (for e.g. Flu Gas Desulfurization tech) should be used in every working coal based thermal power plants.
 - New thermal plants should only be using the latest, most advanced technologies.
 - **Strengthen the EIA** for new plants to consider the impact of Fly Ash and influence of long distance transport of coal.
 - » **Transition to renewable energy** to reduce dependency on fossil fuels
 - » **Promote Energy Efficiency** measures in industries, buildings, households etc. -> this will reduce the need for coal-fired power and lower associated environment impacts.
 - » **Carbon Capture and Storage** to mitigate the contribution of coal-based power generation to climate change.
 - » **Effective water management** practices in coal mining and power generation.
- **Conclusion:** These steps can take us towards reducing the environmental challenges associated with coal-fired thermal plants while facilitating a transition to a more suitable and cleaner energy future.

3) NATIONAL CLEAN AIR PROGRAM: SOLVING AIR POLLUTION PROBLEMS OF INDIAN CITIES

- **Introduction and Problems due to Air Pollution:** Similar to previous topic
- **National Clean Air Program (NCAP)**
 - » NCAP is a flagship initiative of MoEF&CC to fight air pollution. It was launched in Jan 2019 with an aim to reduce PM10 and PM2.5 pollution by at least 20% (20-30%) in five years (i.e. by 2024), with 2017 being the base year. This target was later revised to 40% reduction of PM1.5 and PM10 by 2026.
- **Which are the cities covered?**
 - » **132 Non-attainment Cities** identified on the basis of Ambient Air Quality Data for the period 2011-2015 and WHO report 2014/18.
- **Objectives**
 - » Ensure implementation of **prevention, control and abatement** measures for air pollution
 - » Improve the **monitoring network**.
 - » Enhance **public awareness** regarding air pollution and capacity building measures.
- **Who all are participating?**
 - » Apart from **experts from industry** and academia, **various ministries** like Ministry of Road Transport and Highways, Ministry of Petroleum and Natural Gas, Ministry of New and Renewable Energy, Ministry of Heavy Industries, Ministry of Housing and Urban Affairs, Ministry of Agriculture, Ministry of Health, **NITI Aayog**, and **CBCB** are participating.
- **Key Steps being Taken**
 - i. **Pollution Reduction measures**
 - » Plantation drives, promotion of better technology, sectoral interventions like electric vehicle promotion, promoting renewable energy, waste management etc.
 - » Cities are required to create a Clean Air Action Plan (CAAP) and to prepare this ministry has allocated Rs 10,422 crores.
 - ii. **Enhancing R&D and Data collection**
 - » Studies related to air pollution and its impact are taken up
 - » Monitoring infrastructure is being expanded and will start covering rural areas as well.
 - » A National Emission inventory is to be established to provide proper inputs for future policy making.
 - iii. **Setting up various pollution related institutions**
 - » A National Apex Committee under MoEF&CC, a steering committee under Secretary (environment) and a monitoring committee (under joint secretary) is to be established.
 - » There will be project monitoring committees at the state-level with scientists and trained personnel.
 - » In addition city level review committee under Municipal Commissioner and DM level committee in the districts are to be constituted under NCAP for effective implementation and success of the program.
 - iv. **Increased focus on awareness generation and people's participation.**

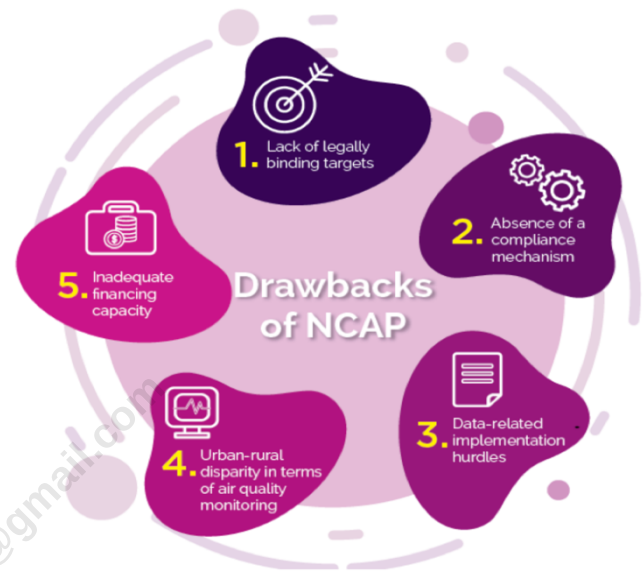
- Analysis: Positives

- » **Kick start of targeted approach:** NCAP has certainly kickstarted the much-awaited good practice of setting air pollution reduction targets.
- » **More comprehensive Action Points** when compared to earlier initiatives. Now the focus should be on strategies of implementation with detailed indicators to enhance the potential impacts.
- » **Health On Board:** NCAP is also supporting health impact studies which was initially missing in draft plan.
- » **Collaborative approach** -> Brings all stakeholders together -> Citizens, scientific community, private sector and government.

- Some other limitations of NCAP

- » **Absence of legal backing** -> NCAP has remained toothless and hasn't been able to implement its action plan
- » **Absence of compliance mechanism** - absence of well-trained people for inspection - > implementation of the initiative has been difficult.
- » **Data related implementation hurdles** - Poor data capture due to substandard monitoring stations; lack of appropriate methodology to leverage real-time data;
- » **Urban-Rural Disparity in terms of air quality monitoring** still is huge. Only a small section of country's population is being monitored.
- » Other experts also criticized **lack of sector wise target, and specific targets for cities**.

Figure 3: Drawbacks of NCAP



- Impact:

» Implementation Delays:

- **Under-usage of funds:** Only 60% of the funds allocated for CAAPs have been used so far.
- **Lack of standard operating procedure (SOPs)** for the implementation process have become an issue.

» Only Marginal Improvement in air quality:

As per the NCAP tracker released in Jan 2023, top polluted non-attainment cities in 2019 have marginally improved their PM 2.5 and PM 10 levels, but continue to breach the CPCB's standard.

- Delhi continues to be the most polluted city
- Almost all the cities continue to breach the CPCB standards.

- Way forward:

- » Despite the above-mentioned hurdles and challenges, the way forward can be accomplished with the right combination of political will, appropriate implementation, a strong compliance mechanism and participation of private players and communities.
- » **Steps government can take:**

- **Improved Infrastructure:** Increasing the number of air pollution monitoring stations (with the ability to measure PM 2.5), especially in rural areas.
 - **Legal Backing** can make the implementation of NCAP and other pollution related initiatives more effective.
 - **More funds for research against air pollution**
 - **Leverage partnerships** with other countries and multilateral organizations for initiatives like technology transfer and knowledge sharing to build a strong framework for combating air pollution.
- » **Utilize India's Cooperative federalism** - Foster greater cooperation and joint action among states. This can take the form of a multi-stakeholder body with representation from the affected states.
- » **Engaging with vulnerable communities:**
- » Representation should be given to women and tribal communities in environmental policymaking and their consent must be necessary for large scale industrial or mining project.
 - » Community based solutions to combat air pollution should be encouraged. These solutions can take the form of community drives for afforestation, SHGs that spread awareness among their members and strengthening capacities of rural communities to adopt cleaner fuels.
- » **How private players can contribute:** Private player is a major contributor to air pollution, and therefore, must be actively engaged to combat air pollution.
- » **An investment fund** with dedicated green focus could play an important role.
 - » **Other ways in which private players can contribute:**
 - » **Cutting edge innovations and technology solutions** - For e.g. Swedish company IKEA uses rice straw as raw material to make furniture - this can be crucial for solving India's stubble burning problem.
 - » **CSR spending** on air pollutions: E.g. Cummins India's initiative of 'Creating Oxygen Hub' where they planted around 35,000 trees with the help of several NGOs, and Maharashtra Forest Department.
 - » **Aligning business operations and supply chains** with clean energy agenda
 - » **Awareness generation** in collaboration with the public sector and other relevant stakeholders.
- **Conclusion2:**
- The journey towards cleaner air in India as charted by NCAP, will be difficult but is necessary. NCAP success hinges on multifaceted approach that combines rigorous scientific studies, strategic funds, and swift and effective implementation of the mitigation methods.

4) DELHI'S AIR POLLUTION PROBLEM

- **Past Year Question:**

- » Mumbai, Delhi and Kolkata are the three mega cities of the country but the air pollution is much more serious problem in Delhi as compared to the other two. Why is this so? [12.5 marks, 200 words] [CSM 2015]

- **Introduction**

- » As per IQAir's **World Air Quality Report, in 2023 Delhi continues to be the worst polluted capital city in the world**. Not only this, it's air quality has worsened in 2023 when compared to 2022.
- » Delhi's air quality dips drastically every year with the arrival of harvest season during October-November. Though, government generally blames the stubble burning in the neighboring states as the key cause, but the air pollution in Delhi is a complex phenomenon that is dependent on a variety of factors. .

- **Key Factors include:**

i. **Input Pollutants**

- » **Dust** is the biggest cause of pollution during winters. **Dry cold weather** leads to dust being easily available in the entire region.
- » **Vehicular Pollution** is the second biggest cause of pollution in winters.
 - According to a study by IIT Kanpur, around 20% of PM2.5 in winters comes from vehicular pollution.
- » **Stubble burning**
 - At the time of Rabi harvesting around 25% of PM content in Delhi is due to stubble burning.
- » **Industries, thermal power plants, burning of waste during winters**
- » **Pollution from firecrackers**

ii. **Meteorological Factors**

- » **Wind Direction:** With the departure of Monsoon, the wind direction in Delhi changes from northeast to northwest
- » **Dip In Temperature** brings the inversion height to lower levels. The concentration of pollutants in the air increases when this happens.
- » **Low Wind speed in winters:** High wind speeds are effective in dispersing pollutants, but with arrival of winters, the average wind speed decreases.

iii. **Other factors**

- » When compared to other metropolitan (i.e. Kolkata, Mumbai and Chennai) - Delhi is surrounded by high density region on all the sides. The other three cities are located near the coast, thus leaving the breathing space for the cities.

- **Steps taken in Delhi in the past to fight Air Pollution:**

- » In 1996, the Supreme Court took a note of the extremely poor air quality of Delhi and ordered the closure and relocation of over 1,300 highly polluting industries from Delhi's residential areas beyond the NCR in a phased manner. Multiple brick kilns were also directed to be relocated outside the city limit.

- » In 1998, the Supreme Court ordered the MoEF&CC to establish an authority for Delhi, which led to the creation of **Environment Pollution Control Authority (EPCA)** in 1998.
 - » Supreme Court subsequently ordered conversion of the whole fleet of DTC buses, taxis and autos to CNG.
 - » During this time centre also took several steps:
 - It revamped its Air Quality Monitoring Program and established a network of monitoring stations, under the National Air Quality Program.
 - » In 2009, National Ambient Air Quality Standards were revised to include the 12 categories of pollutants including PM 2.5 - a noxious pollutant which can penetrate deep into the lungs and even enter the blood stream.
- **Steps taken in recent years to deal with pollution in Delhi**
- » **The Commission for Air Quality Management in National Capital Region and Adjoining Areas**
 - » **Ban on Coal and other unapproved fuels from 1st Jan 2023**
 - » Introduction of **BS-VI** norms for Vehicles
 - » **Graded Response Action Plan (GRAP)**
 - MoEF&CC in early 2017 came out with GRAP.
 - It institutionalized measures to be taken when air quality deteriorates.
 - The plan is **incremental in nature**. The **nature scope and rigor of measures** to be taken is linked to levels of pollution viz. Severe+ or Emergency, Severe, Very Poor, Moderate to Poor and Moderate
 - » **Push for Electric Vehicles - Delhi's Electric Vehicle Policy, 2019** subsidizes buying of electric vehicle and encourage people to move to electrical
 - » **Various steps for controlling stubble burning**
 - » **Prohibition on polluting crackers and promotion of green crackers**
 - » **Odd-Even rule for vehicles** as emergency measures
 - » **Construction of Eastern and Western Peripheral Expressways**
 - To provided fast alternative routes to vehicles not destined for Delhi.
- **Why these steps haven't been able to fight air pollution:**
- » These measures over the years have been fragmented and often reactive.
 - » **Population Growth of Delhi:** Between 1997 - 2011 -> Delhi population increased by 45% and registered cars and two wheelers grew by 250%.
 - » **Focus on pollution control in Delhi's neighborhood** has been minimal -> this negatively impacts the situation in Delhi whose base is already high.
- **Conclusion**
- » Inspite of all the above steps, **improper and incomplete implementation of these initiatives** have led citizens of Delhi to live in a heavily polluted air quality. A coalition of **politics, policy and technology** underpinned by **public will** has to develop to deal with this air pollution problem.

5) THE COMMISSION FOR AIR QUALITY MANAGEMENT IN NCR AND ADJOINING AREAS

- Example Questions

- » How is Commission for Air Quality Management for NCR and Adjoining Areas different from Environment Pollution (Prevention & Control) Authority for the NCR? Will these differences be enough to make the commission for effective? Give reasons [15 marks, 150 words]

- Introduction:

- » The Commission for Air Quality management for NCR and Adjoining Areas is a statutory body formed by an Act of Parliament in Aug 2021. The body was needed as the pollution in the NCR region remains very high and the existing bodies like EPCA, CPCB and SPCBs were found to be ineffective in fighting this pollution problem.
- » It envisages a multi-sectoral, participatory, multi-state dynamic body with a statutory status.

- Differences between the Commission for Air Quality Management for NCR and Adjoining Areas and Environment Pollution (Prevention & Control) Authority for NCR:

- » **Statutory Status:** The commission is backed by a law of parliament, whereas EPCA was a supreme court mandated body. The legal backing gives CAQM more power and authority.
- » **Wider Jurisdiction:** The CAQM has jurisdiction over NCR and Adjoining Areas (state of Punjab, Haryana, Rajasthan and Uttar Pradesh), the EPCA only had jurisdiction over NCR.
- » **Broader Representation:** The commission has representation from Central government, states, NGOs and other expert bodies. The EPCA had much limited membership mainly consisting of court appointed members. The broader representation in CAQM ensures diverse perspectives and better coordination among different stakeholders.
- » **Improved coordination:** The body has the mandate/powers to coordinate among states, which was absent in case of EPCA.
- » **Stronger Enforcement Powers:** CAQM can issues directions to central government, state governments etc. It can impose penalties etc. These powers are much bigger than what EPCA had.

- These differences give the CAQM the ability to be more effective. It has stronger mandate, better coordination ability, and enhance enforcing powers.

- However, the effectiveness of CAQM will depend on:

- i. **Implementation:** CAQM can come with guidelines and orders, but its success will be determined by how effectively its implemented.
- ii. **Political Will** from both central and state government will be crucial in this regard.
- iii. **Improved Infrastructure** for better monitoring and evaluation.
- iv. **Cooperation** between various stakeholders including central government, state governments, local bodies, NGOs etc.

- Conclusion:

- » While the statutory backing, wider jurisdiction and broader representation provides CAQM with opportunities to fight air pollution problems in NCR more effectively, its performance will depend on implementation, political will, better monitoring and increased cooperation among various stakeholders.

6) STRUCTURE, POWER AND FUNCTIONS OF CAQM

- The commission has been set up to monitor and check air pollution levels in the NCR and adjoining region.
 - » It **supersede all existing bodies**. The commission shall have exclusive jurisdiction in respect of matters covered by the law.
 - » **Powers of CPCB/SPCB continue**: The CPCB and its state branches have the power to implement provisions of the Environment Protection Act for air, water and land pollution. However, in case of dispute or clash of jurisdictions, the Commission's writ will prevail specific to matters concerning air pollution.
- **The commission looks at:**
 - » Coordination between states
 - » Planning and execution of policy and interventions
 - » **Research into the cause of pollution**: For e.g. in May 2024, it published the Air Quality Report for NCR Cities/Towns.
 - » Operations of industry
 - » Inspections
 - » **Implementation**: it is empowered to constitute special investigation groups for stricter implementation of air pollution norms on the ground.
 - » **Levy Fine**: It has powers to levy fines ranging upto Rs 1 crore or five years of imprisonment.
- **Structure**
 - The commission is headed by a former or incumbent secretary to the GoI, or chief secretary to a state government. There will be at least six permanent members.
 - Overall there would be **18 members** which would include **five ex-officio members representing the five states**; technical members from CPCB and ISRO; three representatives from NGOs with experience in combating air pollution; one representative from NITI Aayog.
- **NGT**: Only the NGT, and not civil courts, is authorized to hear cases where the commission is involved.
- **Area covered**: Delhi, Punjab, Rajasthan, Haryana and Uttar Pradesh
- **Concerns**
 - » The EPCA had similar powers, but was ineffective in dealing with the air pollution problems of Delhi.
 - » **Too much membership from center** - less membership from states
 - » **Representation from NGOs** also need to increase (currently it's only 3)

7) STUBBLE BURNING AND WAY FORWARD

- **Example Questions**
 - » Discuss the key challenges in controlling the crop stubble burning in states around Delhi? Suggest some remedies to solve the problem of crop stubble burning. [15 marks, 250 words]
 - » "The only solution to this annual stubble burning nightmare lies in changing the economics of the farmer by making collection and disposal of stubble more rewarding than burning" Elaborate [10 marks, 150 words]
- **Introduction:** Every year, in October-November, pollution levels spike in north India. While many factors come together to create smog, **stubble burning remains a significant one**, despite fewer incidents being recorded in 2023 when compared to 2022.
- **Steps Taken to deal with stubble burning so far:**
 - » **Court Orders:** NGT, Courts and Executive Orders banning stubble burning.
 - Increased vigil and fines during stubble burning season.
 - » **'In Situ Management of Crop Residue'** - This is a central scheme under which state governments are given financial support. Under these 50,000 machines have been distributed in Punjab, with **50% subsidy to individual farmers** and **80% subsidy to FPOs**.
 - » **Subsidy for technology** such as happy seeders to reduce stubble burning.
 - » **Awareness generation among farmers:** Various Projects such as **Climate Resilience building among farmers** through crop residue management to increase awareness among farmers.
 - » **Clean Air India** initiative launched to curb air pollution in Indian cities by promoting partnership between Indian and Dutch Companies.
- **Why the stubble burning hasn't stopped** in spite of orders by NGT and Courts?
 - » **Lack of alternative:** Though NGT has banned the stubble burning and governments have also imposed fine farmers seem to have no alternative.
 - **Less time frame** (15-20 days gap between Kharif and Rabi Crop)
 - **Absence of market** for rice straw (no industries)
 - **Straw management equipment** is costly and subsidies are not good enough.
 - **Zero expense** in burning.
- **Why as stubble burning emerged as a key problem over the last few decades?**
 - **Increased mechanization** -> allows simple grain removal, leaving the large stubble
 - **Expansion of paddy cultivation in Haryana, Punjab, Rajasthan and Western UP**
 - » **Subsidy Regime** -> MSP -> farmers eventually moved to rice for assured price and high productivity
 - » **Improvement in irrigation facilities**
 - Rules delaying onset of paddy sowing.
- **Way forward**
 - » **Options of economic benefits from crop residue**
 - Biomass plants could be a possible buyer, and these plants should increase in number

- Compost manufacturing: The stubble can also be used to make compost but has not been explored much yet.
- Manufacturing cardboard, paper etc.
- Some agricultural scientists argue that stubble heaps can be used for mushroom cultivation.
- **Promote cooperative models** for things like biogas plants, compost manufacturing, cardboard or paper manufacturing practices.
- Further, Center should also allow states to include activities like harvesting and composting in MGNREGA.
- A clean air tax could also be imposed, the proceeds of which could be made available to farmers to tackle stubble in a sustainable manner.
- » **Increase the gap between the two crops (rice and wheat)** -> New early maturing varieties
- » **Diversify the crops** being grown in the region.
 - Reforming the subsidy regime: Rationalize electricity cost; Bring UREA under NBS regime; Reform the MSP system to cover more pulse/Oilseed/ and cotton, and less rice and wheat.
 - Incentivization and Institutional Support: Agricultural economists Ashok Gulati suggest Rs 12,000 per hectare (keeping power savings in mind) as an acceptable compensation.
- » **Promoting tools which limits the need of burning**: The current subsidy share needs to increase as farmers currently find this expenditure useless
- » **Explore other technological solutions** like
 - **PUSA Decomposer** - a technology developed by IARI to manage paddy stubble in fields. It can rapidly degrade paddy straws in the field and convert them into compost, which then doesn't pose any issues for sowing of wheat crops, as per IARI scientists.
 - The Swedish technology - **Torrefaction**: This **technology** involves heating up straw, grass, sawmill residue and wood biomass to 250 degrees Celsius - 350 degree Celsius.
- » **Better Monitoring**: ISRO protocol should be adopted to better monitor data of fire incidents related to paddy residue burning. This protocol uses **remote sensing technology**
- » **Long term engagement with farmers**: All the above steps would require that farmers level of awareness, education and income go up
 - Marginal and small farmers should be provided with subsidy to adopt new equipment and technology to solve the stubble burning issue.

- Conclusion

- » The problem of stubble burning is complex and needs a solution which takes into consideration the economic condition of the farmers, the scientific options available and the willingness of the central government to change the policy and fund a major part of the expenditure. Making strong laws, imposing high fines and blaming farmers alone won't solve the problem. Persuading farmers to give up stubble burning requires reaching out to them on multiple fronts, not just with cash incentives.

8) INDIVIDUAL POLLUTANT RELATED TOPICS

A) SULPHURDIOXIDE POLLUTION

- **Example Questions**
 - » Why has India emerged as the largest emitter of SO₂ in the world? Discuss the key ways to tackle the challenge of sulphur pollution. [10 marks, 150 words]
- **Basics**
 - » SO₂ is a colorless gas which has a nasty, sharp smell. It reacts with other substances to form harmful compounds, such as sulfuric acid, sulfurous acid, and sulfate particles.
- **Main Sources** - Burning of Fossil fuels and ships, locomotives using furnace oil/ heavy oil; Other small sources are - **industrial processes** like extracting metal from ore; nature sources such as volcanoes.
- **Why Sulphur dioxide pollution is problematic?**
 - » **Health Issues:** Respiratory diseases; contribute to PM pollution
 - » **Environmental Issues ->**
 - **Harm trees and plants** -> Damaging foliage and decrease growth of trees and plants
 - **ACID Rain** -> harms sensitive ecosystem
 - » **Damage Cultural Heritage**
 - Deposition of sulfur particles may cause discoloration and damage of monuments, statues etc.
 - The fine particles may reduce visibility (Haze)
- **India has emerged as the largest SO₂ emitter in the world: NASA data**
 - » India has **highest number (more than 15%) of all anthropogenic SO₂ hotspots** in the world detected by the **Ozone Monitoring Instrument (OMI)** satellite. These include Singaruli, Neyveli & Chennai, Talcher & Jharsuguda, Korba, Kutch etc.
- **Key reasons for High Sulphur pollution in India**
 - » **Nearly all the SO₂ emission in India** comes from coal burning power plants which are the major source of energy for India.
 - » The vast majority of power plants in India lack flue-gas desulfurization technology to reduce their air pollution.
 - **Note: Flue Gas Desulfurization (FGD)** is a set of technologies used to remove SO₂ from exhaust gas of fossil fuel-based power plants.
- **Recent steps by India**
 - » In Dec 2015, the MoEF&CC had introduced for the first time SO₂ emission limits for coal power plants with an initial deadline to retrofit technology to control emissions from power generations by 2017.
 - This deadline was later **shifted to 2022** and then to 2024
- **Way Forward**
 - » **Incentivize coal-based power plants that comply** with revised pollution norms and **penalize those that don't**
 - This will spur and bring in urgency in the coal-based power sector to abide by the 2015 emission standards.
 - » **Moving to cleaner fuels or renewable options** for energy requirements.

B) AMMONIA POLLUTION

- About Ammonia

- » Ammonia is a colorless gas with characteristics pungent odor. **Natural sources** include decaying organic matter and animal waste. **Manmade sources** include fertilizer manufacturing, waste disposal sites, industrial processes etc.

- Applications

- » 80-90% of ammonia all over the world is used for **making fertilizer** (ammonium nitrate is an important nitrogen fertilizer)
- » It is a precursor of various nitrogenous compounds. Virtually, all synthetic nitrogen compounds are derived from ammonia.
- » It is also used in making household cleaners, plastics, dyes, pharmaceuticals etc.
- » It is an anti-septic and is used in food preservation industry.
- » Scientists are also experimenting with using ammonia as a storage of renewable energy. (Nitrogen gas and water use energy to convert into Ammonia). Later Ammonia can be used a fuel in the fuel cell.

- Ammonia Pollution

- Majority of airborne ammonia comes from fertilizers.
- Ammonia can also contribute to formation of PM2.5 (ammonia combines with VOC, NOx, SO2 etc. to form PM2.5)

- Health Impacts:

- At higher concentration ammonia is toxic, caustic, and hazardous. Exposures at high levels of ammonia can be irritating to a person's skin, eyes, throat, lungs, and cause coughing and burns.
- **Long term health concerns** associated with Ammonia exposure include – severe cardiovascular and respiratory effects, decreased lung function, asthma aggravation, premature death etc.

- **Environmental Impacts** -> Eutrophication, Soil Acidification; biodiversity loss -> promote species which prefer nutrient fueled growth to outcompete other species.

- **Indo-Gangetic Plain Global Hotspot of atmospheric Ammonia: Study by IIT KGP**

- Water Pollution

- » An ammonia concentration of upto 0.5 ppm (BIS) is maximum limit for drinking water. But, if the ammonia concentration is more than 1 ppm, it would negatively impact our health in long run. Similarly, ammonia concentration of more than 1 ppm in water bodies is dangerous for fish population.
- » **High Ammonia Concentration in Yamuna River** is regularly disrupting water supply in Delhi.
 - For e.g., in July 2020, the ammonia concentration in Yamuna River reached 3 ppm. This led to reduction of water supply from Yamuna for Delhi, as Delhi Jal Board doesn't have technology to treat this high concentration of water.
- » **Why ammonia pollution is so high?**

- Industrial units of Sonipat and other drains joining the river along the way may be contributing to this.
- **Way Forward:**
 - » Precision agriculture
 - » Regulating use of fertilizers during different weather conditions
 - » Reduce amount of nitrogen getting into animal waste
 - » **Enhance water treatment technology** -> Ozone-based units can treat ammonia levels upto 4 ppm.
 - Maintain a **sustainable ecological flow** to ensure that the river naturally treats most of the pollutants.

C) FLY ASH

- **Example Questions?**
 - What is Fly Ash? Discuss various environmental and economic aspects associated with it. [10 marks, 150 words]
 - Critically analyze the Draft Fly Ash notification, 2021. What are the different ways in which fly ash can be useful [10 marks, 150 words]
- **Introduction**
 - » Fly ash is a coal combustion byproduct produced in coal based thermal power plants. It refers to ash that is driven out of coal fired boilers together with the flue gases.
 - » In modern coal fired power plants, fly ash is captured by electrostatic precipitators or other particle filtration equipment before the flue gas reaches the chimney.
 - » The composition of fly ash varies considerably, but all fly ash includes substantial amount of Silicon di oxide (SiO_2), Aluminium oxide (Al_2O_3) and Calcium oxide (CaO), the main mineral compounds in coal-bearing rock strata.
- **Key characteristics of fly ash**
 - » **Harmful for human health:** Fly ash contains toxic constituents like lead, cadmium, chromium, arsenic which can be very dangerous for human health. They contribute heavily to particulate matters in air and cause lung problems.
 - » **Bad for environment**
 - Toxic content results in both soil pollution and water pollution (toxic leaching)
 - » **Affects large land area**
 - If fly ash is not captured at the power plant itself, it spreads easily through air and affects large land area.
 - » **It is a pozzolan**, a substance containing aluminous and siliceous material that forms cement in the presence of water. Hence it can be used in construction process.
- **Reducing Fly Ash pollution**
 - » Washing the coal at its place of origin is an important step which ultimately reduces the amount of ash being produced.
 - » Increasing R&D, for enhancing the efficiency of power plants, would also help in reducing the ash content.

- » Capturing fly ash before it is released in air by Chimney using various types of precipitators.
- **Where can the captured fly Ash be used?**
 - » Cement industry uses Fly Ash in the manufacturing of Portland Pozzolana Cement.
 - Recently, scientists at IIT-Hyderabad have found ways to turn fly ash into products like paints, textile coatings etc.
 - » It can also be used fly ash bricks/blocks/ and tiles manufacturing, road embankments construction etc.
 - » Fly ash may also be utilized in agriculture as soil conditioners.
 - » It is also used as a substitute of soil/sand for reclamation of low lying areas.
 - » In mining it can be used for backfilling of mines.
- **Steps taken to promote the use of Fly Ash?**
 - i. **Various notification for fly ash utilization since 1999**
 - The **2016 notification** calls for every agency engaged in construction activity within a radius of 300 km of coal-based thermal power plant to use ash based products for construction.
 - It also mandates the use of ash-based bricks or products in all government schemes and programs.
 - ii. Maharashtra is the first state to have a Fly Ash Utilization Policy. It is also looking to export fly ash to countries like Singapore and Dubai where it is in demand.
 - iii. **GST rates** on fly ash and its products have been reduced to **5%.**
 - iv. Launching of **ASHTrack Mobile App** for better management of fly ash produced by thermal power plant in Feb 2018.
 - It will act as an interface between fly ash producers (thermal power plants) and potential ash users such as road contractors, cement plants etc and thus will help in increasing the utilization of fly ash being produced at coal based thermal power plants.
 - v. **Various awareness campaigns**
 - Through workshops and other programs.
- **Draft Fly Ash Notification (2021)** - it is the proposed 5th amendment to the 1999 fly ash notification.
 - » It introduces a concept of 3-5 years compliance cycle to achieve a target of 100 percent fly ash utilization by the end of cycle.
 - » It also gives an extension of 10 years to power plants to progressively utilize their **legacy ash.**
- **Conclusion**
 - » Coal is going to remain the mainstay of power for some years to come. Further, in India coal has much more ash content when compared to coal from other countries. Therefore we should take all steps possible to reduce the ash being produced and utilize the ash which actually gets produced. This should include ensuring compliance with the Fly Ash notifications; bringing in more stringent norms; increasing awareness about the use of fly ash and prioritizing the proper disposal/usage of legacy ash.

D) NITROGEN POLLUTION

- Example Questions

- » While carbon has been debated upon extensively by the Inter-governmental Panel on Climate Change and under the United Nations Framework Convention on Climate Change, it is time for us to look at N₂O and nitrogen pollution in general in a big way. Discuss [15 marks, 250 words]

- Introduction

- » While nitrogen is the dominant gas in the atmosphere, it is inert and doesn't react. However, when it is released as part of compounds from agriculture, sewage and biological waste, nitrogen is considered 'reactive' and may be polluting and causing greenhouse effect.
- » The release of these reactive nitrogen compounds in the atmosphere have increased over the years because of increased use of **fertilizers for agriculture** and increased **industrial pollution**. NO_x emissions grew at 52% from 1991 to 2001 and 69% from 2001-2011.
- » In fact, a study in 2017 showed that we have **breached the planetary boundary of N (Nitrogen)**. This planetary boundary is set at 44 Tg (Tera-grams) per year globally. But, currently we use 150 Tg N per year, primarily through fertilizer usage.

- Key Highlights

- » **Human emission** of N₂O increased 30% in 36 years.
- » 43% of the total emissions came from human sources.
- » The increase in emission means that climate burden from non-carbon sources is also increasing.
- » **Dichotomy between Climate Crisis and Food Security** - Major proportion of the N₂O emissions in the last four decades came from the agricultural sector, mainly because of the use of nitrogen-based fertilizers.
- » **Most of the emission** have come from **developing countries** like China, India and Brazil.

- Key causes of Nitrogen Pollution

- » **Emission from chemical fertilizer** - about 50% of the nitrogen used in global agri sector is released in environment (atmosphere, water bodies etc.). Most important source; difficult to control (non point source, food security concerns etc.);
- » **Sewage and organic solid wastes** (second largest source):
 - The poor systems of sewage and solid waste management and poor recovery of nutrients has made it a key concern in India.
- » **Burning of fossil fuels**
 - Vehicular pollution, mostly from road transport is another major NO_x producer.
- » **Industries**

- Key threats due to nitrogen pollution

- » **Air Pollution:** Emissions of Ammonia, nitrogen oxide and nitrous oxide contribute to **particulate matter and acid rain**. These cause respiratory problems and cancers for people and damage to forests and buildings.
- » **Water Pollution - Eutrophication**
- » **Negatively hampers soil health** -> brings down the yield of agri-land

- » **Climate Change:** Nitrous Oxide (N₂O) -> GWP: 300 times of CO₂; Also contributes to Ozone depletion.
- » **Health, economy and livelihood**
 - Deteriorating soil quality impacts Agri output and livelihood. Further, particulate matter and acid rains have adverse impact on health.
- **UNEP's Colombo Declaration on Sustainable Nitrogen Management (Oct 2019)**
 - » Sri Lanka, with support from the UNEP, convened an event at which member states came together to adopt what is called the "**Colombo Declaration**".
 - » **Key Highlights**
 - Halve nitrogen waste by 2030.
 - The member countries also endorsed UN's plan for a sustainable nitrogen management called "**Nitrogen for Life**", which stems from the Sustainable Nitrogen Management Resolution which was adopted during the fourth session of the UN environment Assembly held from 11-15th March 2019, at the UNEP headquarter in Nairobi.
 - » **Analysis**
 - This is the first time governments have agreed to work together on a major quantitative global goal for improved nitrogen management.
- **Way Forward - will be done for mains**
 - » **Optimize the use of nitrogen and minimize the losses**
 - **Rationalize the unbalanced use of nitrogen fertilizers** in irrigated areas that grow multiple crops in a year and ensure need-based usage of nitrogen fertilizer in all areas.
 - Move to **low nitrogen foot print diet** such as vegetables, chicken, sea food etc.
 - **Reduce food waste** -> improve FCI storage facilities, optimal consumptions/utilization by individual citizens.
 - **Well-established practices and technologies** to mitigate nitrous oxide pollution do exist and industrial and agricultural policies in Europe have reduced the emissions considerably.
 - » **Technology** (e.g. moving to BS-VI norms)
 - To modify industrial systems and vehicle engines -> reduced NO_x emissions.
 - » **Policy**
 - Government needs to focus on reducing subsidy on UREA and promoting balanced fertilizers. Further, procurement policies have to be modified, infrastructure has to be created to reduce the food wastage.
 - » **Consumer Action**
 - Citizens can also contribute immensely towards reducing nitrogen pollution. They can put pressure on government, reduce food wastage and thus promote optimum utilization.

E) WHO'S AIR QUALITY GUIDELINES

- In Sep 2021, WHO revised the air quality guidelines. This was the first major update to the standards in 15 years.
- **Why was there a need of update?**
 - New studies have found that even smaller quantity of pollutants were harmful for human beings.
- Therefore, WHO has strengthened nearly all pollutant standards in comparison to the quality guidelines established in 2005 (published in 2006)
 - **Expected Impact:** If the target levels are implemented and achieved by governments, it would lead to saving of lakhs of lives.
- **The new guidelines recommend air quality levels for 6 pollutants**, where evidence has advanced the most on health effects from exposure.

Recommended 2021 AQG levels compared to 2005 air quality guidelines

Pollutant	Averaging Time	2005 AQGs	2021 AQGs
PM _{2.5} , µg/m ³	Annual	10	5
	24-hour ^a	25	15
PM ₁₀ , µg/m ³	Annual	20	15
	24-hour ^a	50	45
O ₃ , µg/m ³	Peak season ^b	-	60
	8-hour ^a	100	100
NO ₂ , µg/m ³	Annual	40	10
	24-hour ^a	-	25
SO ₂ , µg/m ³	24-hour ^a	20	40
CO, mg/m ³	24-hour ^a	-	4

- **Expected impact of the new guidelines:**
 - » Spur greater global reactions in pollution emissions.
 - » Contribute to fight against climate.
- **Note:** These guidelines are not legally binding on any country. But, countries and legislative bodies regularly refer to WHO guidelines when setting airborne pollutant control legal policy.
- **Implications for India**
 - As per the new WHO guidelines, almost the entire India, specially the Urban areas would now be considered polluted for entire year.

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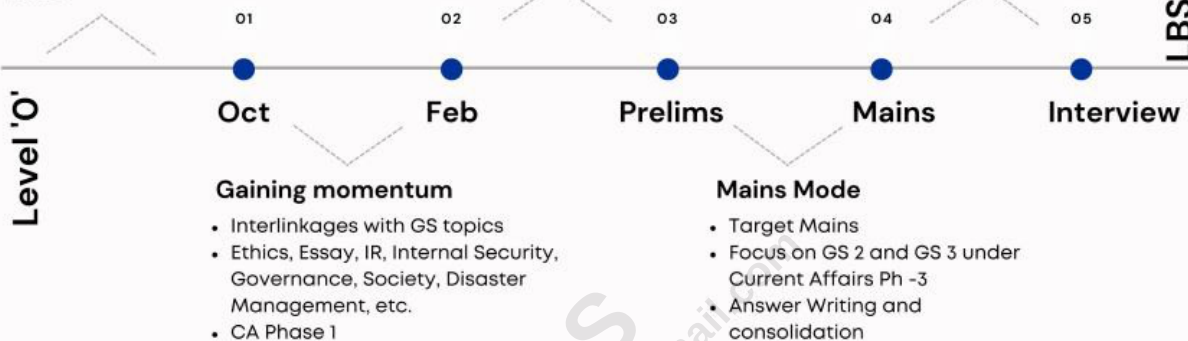
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